

Sequences & Series



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12} .
 - 2 For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15} .
 - 3 A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.
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Arithmetic series (sum formula)

- 4 Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).
 - 5 Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.
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Geometric sequences

- 6 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6 .
 - 7 A geometric sequence begins 5, -10 , 20, -40 , Find r and a_7 .
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Geometric series (finite)

- 8 Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).
 - 9 Find S_5 for the geometric sequence 5, -10 , 20, -40 ,
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Infinite geometric series

- 10 Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \dots$.
 - 11 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.
 - 12 Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.
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Multiple choice — easy

13 For the arithmetic sequence with $a_1 = -15$ and common difference $d = 8$, find a_{16} .

a. 113

b. 105

c. 97

d. 107

14 For the arithmetic sequence with $a_1 = 19$ and common difference $d = 6$, find a_{11} .

a. 73

b. 81

c. 79

d. 85

15 For the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$, find a_{21} .

a. -74

b. -78

c. -76

d. -82

16 For the arithmetic sequence with $a_1 = -15$ and common difference $d = -5$, find a_{19} .

a. -110

b. -103

c. -100

d. -105

- 17 For the arithmetic sequence with $a_1 = 18$ and common difference $d = -3$, find a_{18} .
- a. -36
 - b. -33
 - c. -31
 - d. -30
- 18 For the arithmetic sequence with $a_1 = 3$ and common difference $d = -5$, find a_{20} .
- a. -97
 - b. -90
 - c. -92
 - d. -87
- 19 For the arithmetic sequence with $a_1 = 10$ and common difference $d = 7$, find a_{15} .
- a. 108
 - b. 110
 - c. 101
 - d. 115
- 20 For the arithmetic sequence with $a_1 = -9$ and common difference $d = -1$, find a_{13} .
- a. -19
 - b. -20
 - c. -21
 - d. -22

- 21 For the arithmetic sequence with $a_1 = 16$ and common difference $d = 8$, find a_{13} .
- a. 114
 - b. 120
 - c. 104
 - d. 112
- 22 For the arithmetic sequence with $a_1 = 17$ and common difference $d = 5$, find a_{15} .
- a. 82
 - b. 87
 - c. 92
 - d. 89
- 23 For the arithmetic sequence with $a_1 = -5$ and common difference $d = -8$, find a_{21} .
- a. -163
 - b. -157
 - c. -165
 - d. -173
- 24 For the arithmetic sequence with $a_1 = -18$ and common difference $d = 6$, find a_{19} .
- a. 92
 - b. 96
 - c. 90
 - d. 84

- 25 For the arithmetic sequence with $a_1 = 19$ and common difference $d = -6$, find a_{15} .
- a. -71
 - b. -59
 - c. -63
 - d. -65
- 26 For the arithmetic sequence with $a_1 = 3$ and common difference $d = -4$, find a_{21} .
- a. -75
 - b. -81
 - c. -73
 - d. -77
- 27 For the arithmetic sequence with $a_1 = -17$ and common difference $d = -4$, find a_{16} .
- a. -81
 - b. -75
 - c. -77
 - d. -73

Multiple choice — medium

- 28 For the geometric sequence with $a_1 = 3$ and common ratio $r = -2$, find a_7 .
- a. 192
 - b. -384
 - c. 190
 - d. 194

- 29 For the geometric sequence with $a_1 = 4$ and common ratio $r = -3$, find a_7 .
- a. -8748
 - b. 2913
 - c. 2916
 - d. 2919
- 30 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_7 .
- a. 8748
 - b. 2913
 - c. 2916
 - d. 2919
- 31 For the geometric sequence with $a_1 = -1$ and common ratio $r = 2$, find a_6 .
- a. -64
 - b. -32
 - c. -30
 - d. -34
- 32 For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_7 .
- a. -192
 - b. -190
 - c. -194
 - d. -384

- 33 For the geometric sequence with $a_1 = -1$ and common ratio $r = 2$, find a_4 .
- a. -8
 - b. -10
 - c. -6
 - d. -16
- 34 For the geometric sequence with $a_1 = 1$ and common ratio $r = -3$, find a_8 .
- a. -2184
 - b. 6561
 - c. -2190
 - d. -2187
- 35 For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_6 .
- a. -30
 - b. 64
 - c. -32
 - d. -34
- 36 For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_4 .
- a. -8
 - b. -10
 - c. 16
 - d. -6

- 37 For the geometric sequence with $a_1 = -3$ and common ratio $r = -3$, find a_6 .
- a. 726
 - b. 729
 - c. 732
 - d. -2187
- 38 For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_8 .
- a. -386
 - b. -384
 - c. -382
 - d. -768
- 39 For the geometric sequence with $a_1 = 1$ and common ratio $r = 3$, find a_6 .
- a. 240
 - b. 246
 - c. 243
 - d. 729
- 40 For the geometric sequence with $a_1 = 2$ and common ratio $r = 3$, find a_6 .
- a. 483
 - b. 1458
 - c. 486
 - d. 489

- 41 For the geometric sequence with $a_1 = 3$ and common ratio $r = -3$, find a_6 .
- a. -726
 - b. -732
 - c. -729
 - d. 2187
- 42 For the geometric sequence with $a_1 = 4$ and common ratio $r = -2$, find a_5 .
- a. -128
 - b. 64
 - c. 62
 - d. 66
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Multiple choice — hard

- 43 Find S_{13} for the arithmetic sequence with $a_1 = -8$, $d = 1$.
- a. -27
 - b. -26
 - c. -13
 - d. -4
- 44 Find S_{13} for the arithmetic sequence with $a_1 = 10$, $d = 8$.
- a. 116
 - b. 754
 - c. 746
 - d. 767

45 Find S_{12} for the arithmetic sequence with $a_1 = 2$, $d = 9$.

a. 609

b. 630

c. 618

d. 103

46 Find S_{12} for the arithmetic sequence with $a_1 = -1$, $d = 4$.

a. 42

b. 252

c. 248

d. 264

47 Find S_{12} for the arithmetic sequence with $a_1 = 7$, $d = -9$.

a. -510

b. -85

c. -501

d. -498

48 Find S_4 for the geometric sequence with $a_1 = 4$, $r = 3$.

a. 163

b. 157

c. 108

d. 160

49 Find S_4 for the geometric sequence with $a_1 = 2$, $r = -2$.

a. -10

b. -16

c. -8

d. -12

50 Find S_4 for the geometric sequence with $a_1 = 4$, $r = -3$.

a. -77

b. -108

c. -80

d. -83

51 Find S_6 for the geometric sequence with $a_1 = 6$, $r = -2$.

a. -126

b. -192

c. -124

d. -128

52 Find S_6 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -1095

b. -1092

c. -1089

d. -729

- 53 Find the sum of the infinite geometric series with $a_1 = 13$ and $r = \frac{1}{2}$.
- a. 26
 - b. $\frac{26}{3}$
 - c. $\frac{13}{2}$
 - d. 27
- 54 Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.
- a. 21
 - b. $\frac{7}{3}$
 - c. $\frac{21}{5}$
 - d. $\frac{21}{2}$
- 55 Find the sum of the infinite geometric series with $a_1 = 4$ and $r = -\frac{1}{2}$.
- a. 8
 - b. $\frac{8}{3}$
 - c. 6
 - d. -8
- 56 Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.
- a. 27
 - b. 3
 - c. $\frac{27}{5}$
 - d. $\frac{27}{2}$

57 Find the sum of the infinite geometric series with $a_1 = 18$ and $r = \frac{1}{5}$.

a. 90

b. $\frac{45}{2}$

c. 15

d. $\frac{72}{5}$